



Motorsport

USER MANUAL
BMW M HYBRID V8 LMDH



Motorsport

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Motorsport

DEAR iRACING USER,

Congratulations on your purchase of the BMW M Hybrid V8 LMDh! From all of us at iRacing, we appreciate your support and your commitment to our product. We aim to deliver the ultimate sim racing experience, and we hope that you'll find plenty of excitement with us behind the wheel of your new car!

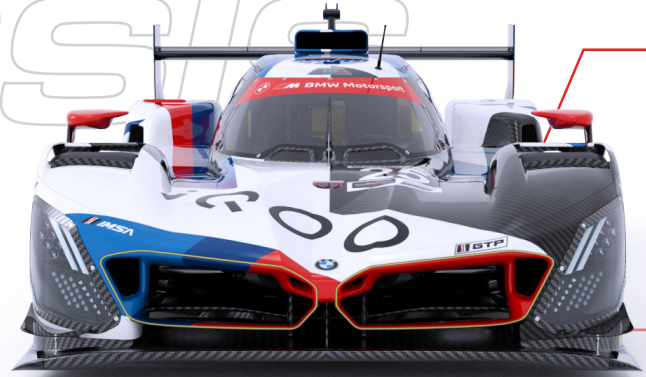
The following guide explains how to get the most out of your new car, from how to adjust its settings off of the track to what you'll see inside of the cockpit while driving. We hope that you'll find it useful in getting up to speed.

Thanks again for your purchase, and we'll see you on the track!



CHASSIS

**PUSHROD-ACTUATED INDEPENDENT
TORSION BAR FRONT AND REAR
SUSPENSION WITH FRONT HEAVE AND
REAR THIRD-SPRING ELEMENTS**



LENGTH
5100 mm
200.8 in

WIDTH
2000 mm
78.7 in

WHEELBASE
3148 mm
124 in

DRY WEIGHT
1030 kg
2270 lbs

WET WEIGHT
WITH DRIVER
1150 kg
2535 lbs

POWER
UNIT

**CAST ALUMINUM V8 WITH
HYBRID MGU-K RECOVERY
SYSTEM**

DISPLACEMENT
4.0 Liters
146.5 CID

RPM LIMIT
8200 RPM

TORQUE
480 lb-ft
650 Nm

POWER
640 bhp
477 kW



INTRODUCTION

The information found in this guide is intended to provide a deeper understanding of the chassis setup adjustments available in the garage, so that you may use the garage to tune the chassis setup to your preference.

Before diving into chassis adjustments, though, it is best to become familiar with the car and track. To that end, we have provided baseline setups for each track commonly raced by these cars. To access the baseline setups, simply open the Garage, click iRacing Setups, and select the appropriate setup for your track of choice. If you are driving a track for which a dedicated baseline setup is not included, you may select a setup for a similar track to use as your baseline.

After you have selected an appropriate setup, get on track and focus on making smooth and consistent laps, identifying the proper racing line and experiencing tire wear and handling trends over a number of laps.

Once you are confident that you are nearing your driving potential with the included baseline setups, read on to begin tuning the car to your handling preferences.

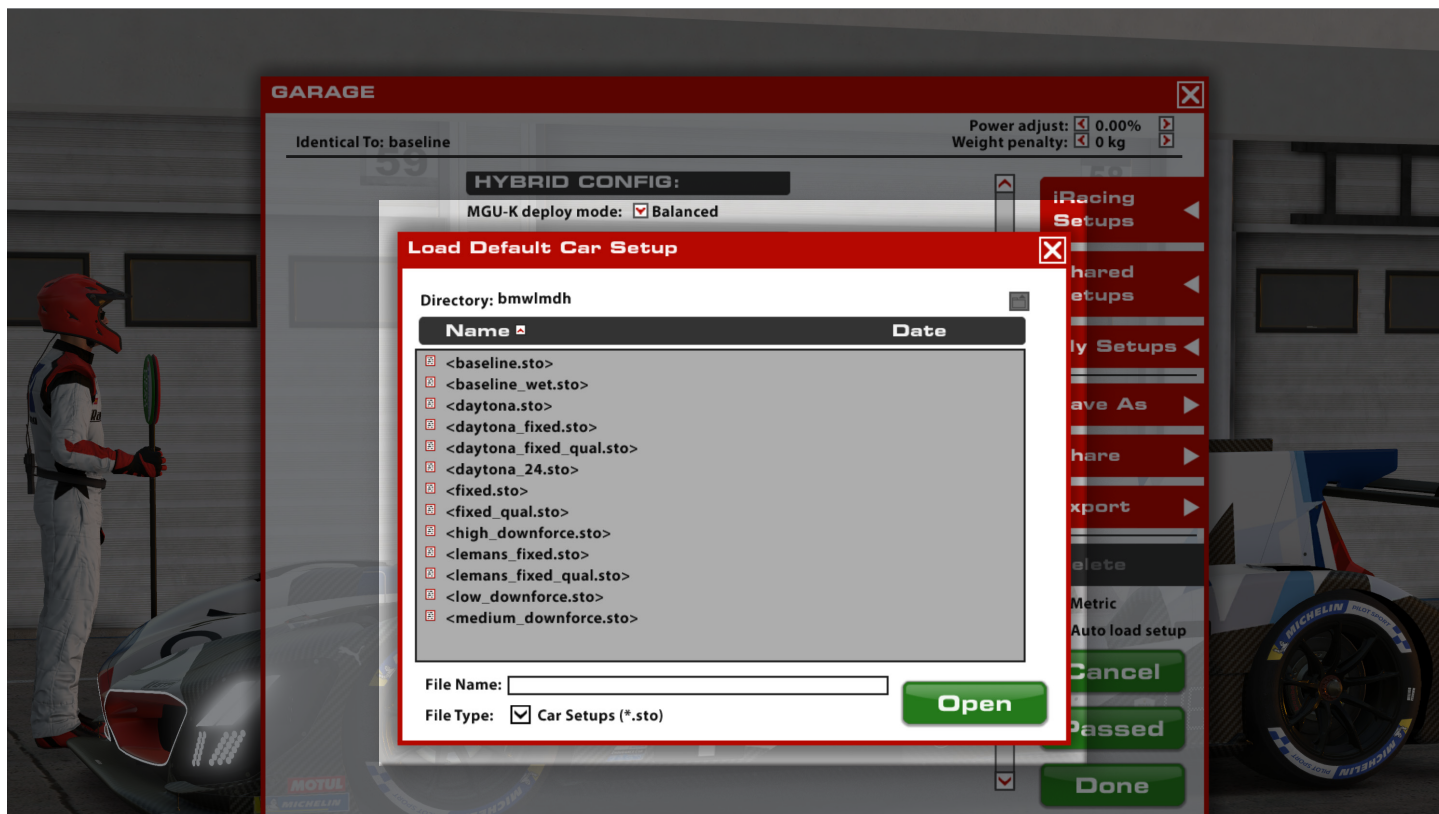
GETTING STARTED



Before starting the car, it is recommended to map controls to adjust the Brake Bias and Traction Control systems. While this is not mandatory to drive the car, this will allow you to make quick changes to the driver aid systems to suit your driving style while out on the track.

Once you load into the car, getting started is as easy as selecting the “upshift” button to put it into gear, and hitting the accelerator pedal. This car uses a sequential transmission and does not require a clutch input to shift up or down. However, the car’s downshift protection will not allow you to downshift if it feels you are traveling too fast for the gear selected and would incur engine damage. In these situations the downshift command will simply be ignored. Upshifting is recommended when all of the shift lights on the steering wheel have changed to red.

LOADING AN iRACING SETUP



Upon loading into a session, the car will automatically load the iRacing Baseline setup [baseline.sto]. If you would prefer one of iRacing's pre-built setups that suit various conditions, you may load it by clicking Garage > iRacing Setups > and then selecting the setup to suit your needs.

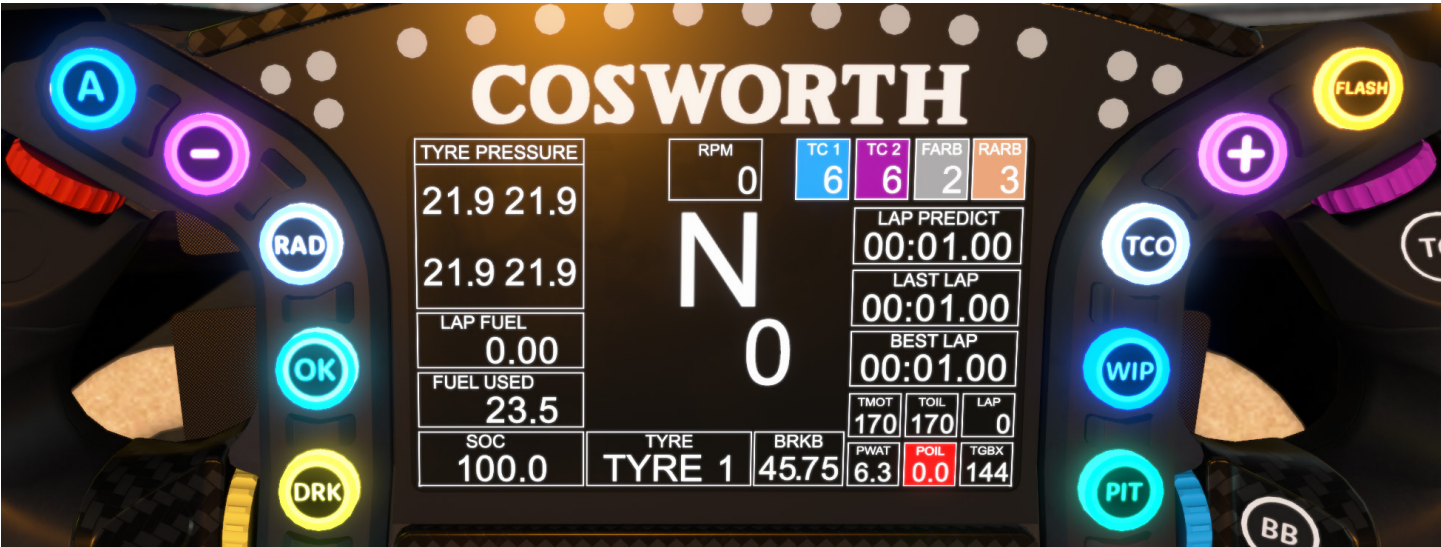
If you would like to customize the setup, simply make the changes in the garage that you would like to update and click apply. If you would like to save your setup for future use click "Save As" on the right to name and save the changes.

To access all of your personally saved setups, click "My Setups" on the right side of the garage. If you would like to share a setup with another driver or everyone in a session, you can select "Share" on the right side of the garage to do so.

If a driver is trying to share a setup with you, you will find it under "Shared Setups" on the right side of the garage as well.

DASH CONFIGURATION

Vital information relating to the car's performance and driver-adjustable settings can all be found on the digital display integrated into the BMW M Hybrid V8 LMDh's steering wheel.



Left Column

Tyre Pressure	Current air pressure in each tire, updated live and displayed in either PSI or Bar.
Lap Fuel	The amount of fuel used in the previous lap, in either gallons or liters, is updated at the end of each lap.
Fuel Used	The amount of fuel that has been used, in gallons or liters, since the fuel was last added to the car.
SOC	Current battery charge level, in percent.

Center Column

RPM	The current engine RPM is shown at the top of the center section
Gear Indicator	The currently-selected gear is shown as a large number in the center of the display.
BEAM	If the car's nighttime light system is on, this indicator will appear in the lower left of the center area.
Tyre	Inoperable
BRKB	Inoperable

Right Column

TC1	Displays the current Traction Control Gain setting
TC2	Displays the current Traction Control Slip setting
FARB	Displays the current setting for the Front Anti-Roll Bar
RARB	Displays the current setting for the Rear Anti-Roll Bar
Lap Predict	The onboard systems will attempt to predict the current lap time based on splits and performance and display it in the top right.
Last Lap	The previously completed lap time, updated and displayed at the end of each lap
Best Lap	The fastest lap of the current session
TMOT	The engine's cooling fluid temperature, in either °F or °C
TOIL	Engine oil temperature, in either °F or °C
LAP	Displays the number of laps completed in the current session
PWAT	Engine cooling fluid pressure, in PSI or Bar
POIL	Engine Oil system pressure, in PSI or Bar
TGBX	Gearbox oil temperature, in °F or °C



WHEELSPIN / BRAKE LOCK INDICATOR LIGHTS

Two sets of LED lights are situated on the dash to quickly communicate to the driver if any wheelspin or brake lockup is occurring. The left LED cluster corresponds to the left wheels and the right cluster corresponds to the right wheels.



Traction Control

When the clusters light up all LEDs in blue the Traction Control system is intervening to reduce wheelspin. Severe cases of wheelspin and TC system activation will produce a solid blue light and the lights will begin flashing for less amounts of TC intervention.



Brake Lockup

Whenever braking force is sufficient to begin locking a wheel the LEDs will illuminate to show both which wheel is locking and how severe the lockup is. Pink lights indicate front wheel lockup and yellow lights indicate rear wheel lockups, with lockups becoming more severe with more LEDs illuminated. The image above shows both front wheels locked with a slight lockup on the left-rear wheel.

SHIFT LIGHTS



The top of the steering wheel has a set of lights to help the driver know when to shift up to the next gear while accelerating. As RPM increases, the lights will illuminate from the left to right starting with green and ending with two red LEDs on the right.



Once the optimum shift point has been reached all LEDs will change to red and begin flashing.

PIT LIMITER



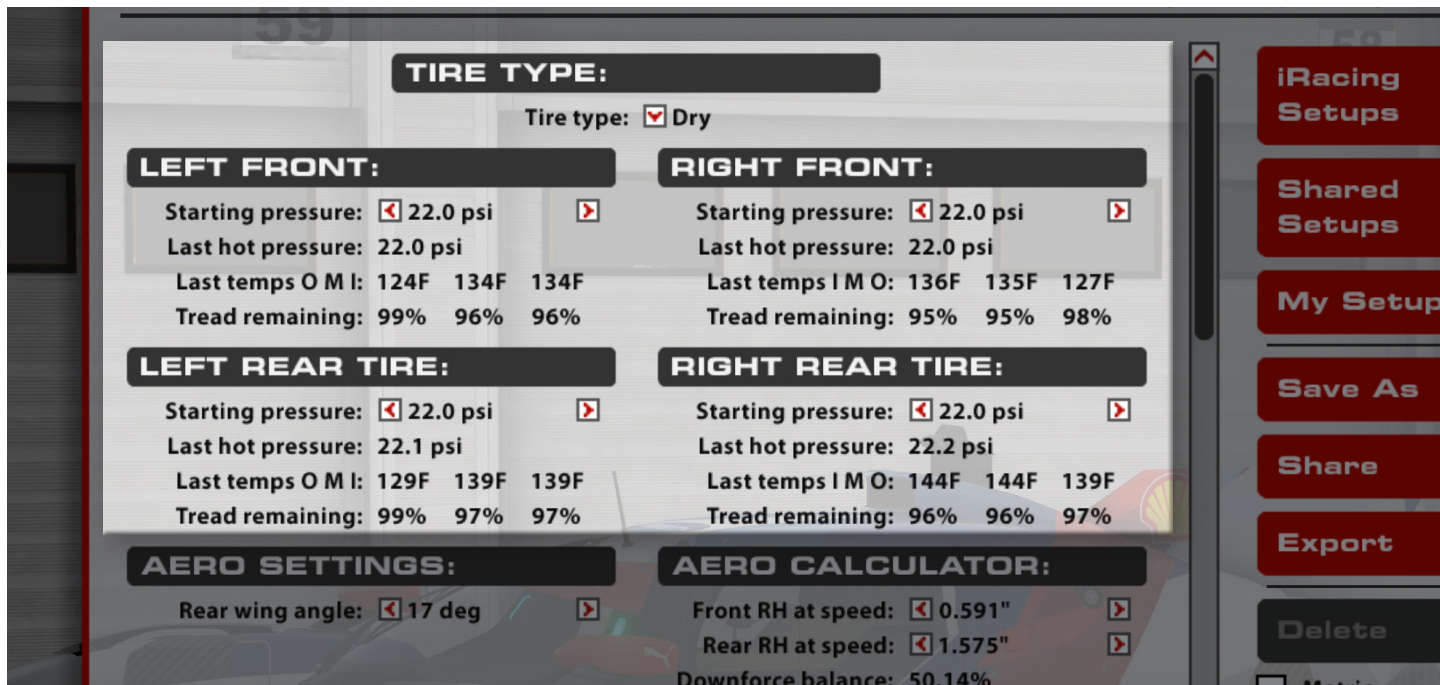
Whenever the Pit Speed Limiter is active the dash will feature a magenta box next to the gear indicator with PSL displayed

ADVANCED SETUP OPTIONS

This section is aimed toward more advanced users who want to dive deeper into the different aspects of the vehicle's setup. Making adjustments to the following parameters is not required and can lead to significant changes in the way a vehicle handles. It is recommended that any adjustments are made in an incremental fashion and only singular variables are adjusted before testing changes.

TIRES & AERO

TIRE DATA



TIRE TYPE

Selects which type of tire is installed on the car when loaded into the world. Dry, or slick, tires are used for dry racing conditions while Wet tires are intended for raining and wet track conditions.

STARTING PRESSURE

The air pressure in the tires when the car is loaded into the world. Lower pressures will provide more grip but will produce more rolling drag and build temperature faster. Higher pressures will feel slightly more responsive and produce less rolling drag, but will result in less grip. Generally, higher pressures are preferred at tracks where speeds are higher while lower pressures work better at slower tracks where mechanical grip is important.

LAST HOT PRESSURE

When the car returns to the garage after an on-track stint, the tire pressure will be displayed as Hot Pressure. The difference between cold and hot pressure is a good way to see how tires are being loaded and worked while on track. Tires seeing more work will build more pressure, and paying attention to which tires are building more pressure and adjusting cold pressure to compensate can be crucial for optimizing tire performance.

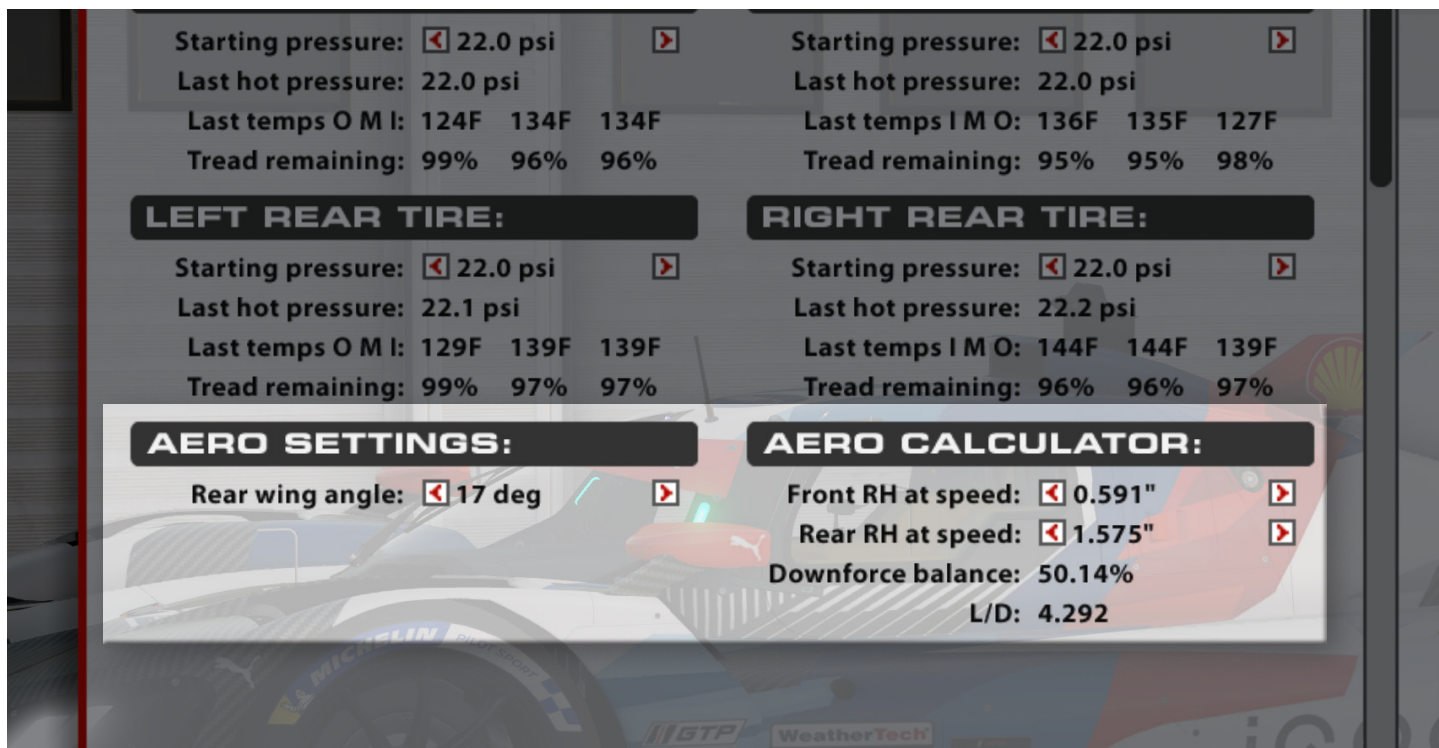
LAST TEMPS

The tire carcass temperatures (measured within the tread) are displayed after the car returns from the track. These temperatures are an effective way to determine how much work or load a given tire is experiencing while on track. Differences between the inner and outer temperatures can be used to tune individual wheel alignment and the center temperatures can be compared to the outer temperatures to help tune tire pressure.

TREAD REMAINING

The amount of tread on the tire, displayed as a percentage of a new tire, is shown below the tire temperatures. These values are good for determining how far a set of tires can go before needing to be replaced, but don't necessarily indicate an under- or over-worked tire in the same way temperatures will.

AERO SETTINGS



REAR WING ANGLE

The Rear Wing Angle setting changes the angle of attack of the rear wing. Increasing the angle will increase the amount of downforce produced and move the aero balance rearward but will increase the amount of drag the wing produces. Reducing the angle will reduce overall downforce and shift aero balance forward, but will reduce drag and allow for a higher top speed. This angle is expressed as a reference measurement, not an absolute angle relative to the ground.

AERO CALCULATOR

The Aero Calculator is a tool used to display the car's approximate aerodynamic values in a given configuration. Changes to the car's aerodynamic settings will be reflected in the Aero Calculator, giving an idea of how the car will behave aerodynamically while on the race track. This calculator can also be used to determine what changes need to be made to the car to alleviate aerodynamically-induced handling issues.

FRONT/REAR RH AT SPEED

The Ride Height (RH) at Speed is used to give the Aero Calculator heights to reference for aerodynamic calculations. When using the aero calculator, determine the car's Front and Rear Ride height via telemetry at any point on track and input that value into the "Front RH at Speed" setting.

DOWNFORCE BALANCE

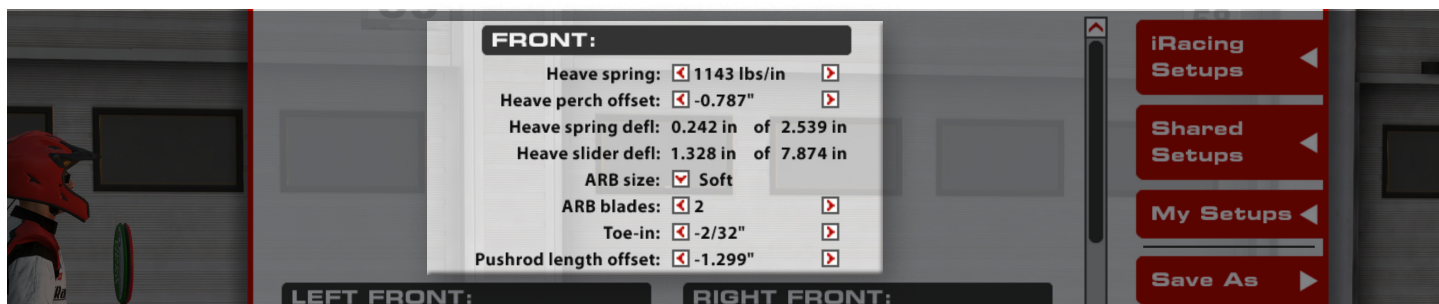
Displayed in percent of Front downforce, this value shows how much of the car's total downforce is over the front axle. A higher percentage value indicates more front downforce, increasing oversteer in mid- to high-speed corners and a lower percentage value indicates more rear downforce, increasing understeer in mid- to high-speed corners.

L/D

The "L/D" value is the ratio of Lift (downforce) to Drag. This quantifies how efficiently the car's bodywork is producing downforce in terms of how much drag is being produced as a result. A higher L/D value means more downforce is being produced for each unit of drag, meaning the bodywork is being more efficient. Having a higher L/D value without sacrificing overall downforce will result in a faster, more efficient car. Optimum values for L/D can vary based on the aerodynamic configuration and track type.

CHASSIS

FRONT



HEAVE SPRING

The front Heave Spring is a suspension element that handles external loads from purely vertical loads and doesn't control loads that would induce chassis roll when cornering. Generally these loads are present for increasing downforce loads at higher speeds, dips and crests in the track, or under heavy braking. Higher rate values will stiffen the suspension in heave, which is good for controlling ride heights to maintain a good aerodynamic platform, but can produce a bouncing effect on rough surfaces. Lower rates will absorb bumps and loads easier, but will hurt the aerodynamic platform due to excessive chassis movement.

HEAVE PERCH OFFSET

The Heave Perch Offset is used to adjust preload on the Heave Spring. This is one of two methods to adjust ride height through the front Heave element, with lower values preloading the spring more and raising front ride heights. Conversely, higher values will unload the spring and lower front ride heights.

HEAVE SPRING DEFLECTION

Heave Spring Deflection represents the amount the Heave Spring is compressed under static conditions. This is not directly adjustable but will change with adjustments to the Heave Perch Offset and front Torsion Bar settings.

HEAVE SLIDER DEFLECTION

The Slider Deflection is how far the slider mechanism the Heave Spring is mounted on has compressed from fully extended. Similar to a shock but without any damping forces produced, this doesn't influence the suspension's behavior.

ARB SIZE

The ARB (Anti-Roll Bar) size alters the stiffness of the front suspension in roll. Increasing the ARB size will increase the roll stiffness of the front suspension, resulting in less body roll but increasing mechanical understeer. This can also, in some cases, lead to a more responsive steering feel from the driver. Conversely, reducing the ARB size will soften the suspension in roll, increasing body roll but decreasing mechanical understeer. This can result in a less-responsive feel from the steering, but grip across the front axle will increase.

ARB BLADES

The configuration of the Anti-Roll Bar arms, or "blades", can be changed to alter the overall stiffness of the ARB assembly. Higher values transfer more force through the arms to the ARB itself, increasing roll stiffness in the front suspension and producing the same effects, albeit on a smaller scale, as increasing the diameter of the sway bar. Conversely, lower values reduce the roll stiffness of the front suspension and produce the same effects as decreasing the diameter of the sway bar. These blade adjustments can be thought of as fine-tuning adjustments between sway bar diameter settings.

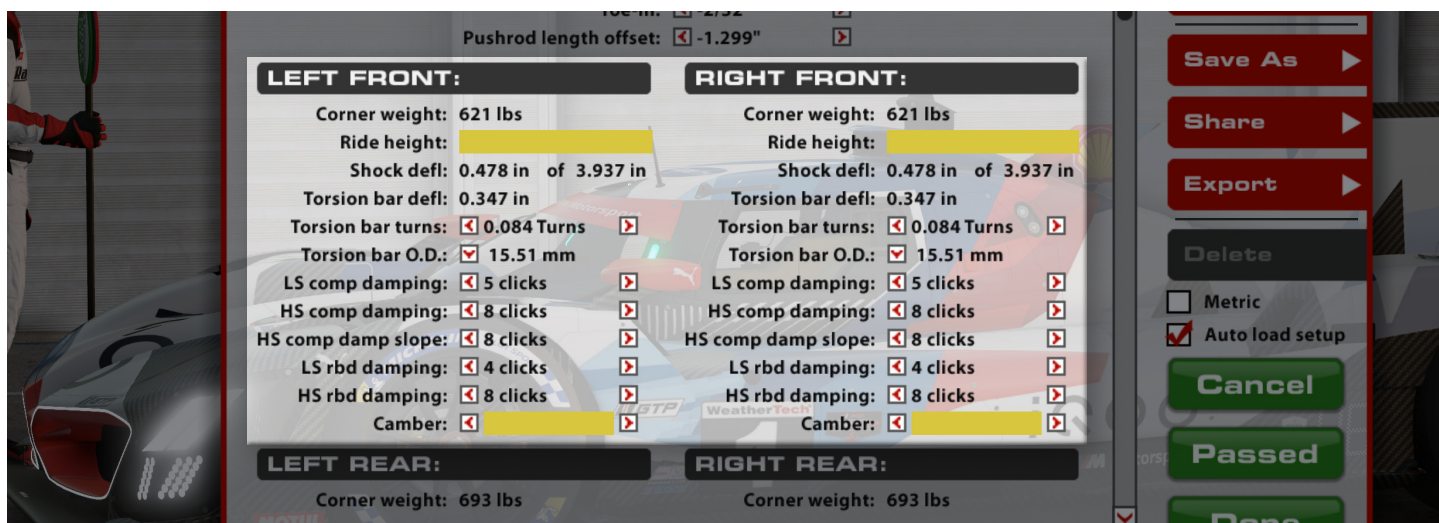
TOE-IN

Toe is the angle of the wheel, looking from vertical, relative to the chassis centerline. Toe-in is when the front of the wheels are closer to the centerline while Toe-out is when the front of the wheels are farther from the centerline than the rear of the tires. On the front end, Toe will alter how quickly the tires respond to steering inputs and influence how stable the car is in a straight line. Toe-out settings (negative garage value) will increase turn-in response and make the car less stable in a straight line, while Toe-in (positive garage value) will increase straight-line stability while making initial steering response more sluggish.

PUSHROD LENGTH OFFSET

This adjusts the length of both front suspension pushrods together, shown as an offset from a baseline length figure. This is a great way to adjust front ride height without altering the preload on the Heave Spring or either front Torsion Bars.

FRONT CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Torsion Bar Turns setting on the front corners and the Spring Perch Offset on the rear corners.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Increasing the front ride height will decrease overall downforce and shift the aerodynamic balance rearward, but will decrease drag slightly. Conversely, reducing front ride height will increase downforce and shift aero balance forward while slightly increasing overall drag.

SHOCK DEFLECTION

EShock Deflection is how much the shock has compressed from its fully extended length while under static conditions in the garage. This is useful for determining how much shock travel is available before a bump stop is engaged on the shock.

TORSION BAR DEFLECTION

The Torsion Bar Deflection is a representation of how much the front torsion bar springs have been preloaded under static conditions. Higher deflection values show higher amounts of spring preload, lower values represent less spring preload.

TORSION BAR TURNS

Used to adjust ride height and corner weight, adjusting this setting applies a preload to the torsion bar under static conditions. Decreasing the value increases preload on the torsion bar, adding weight to its corner and increasing the ride height at that corner. Increasing the value does the opposite, reducing height and weight on a given corner. These should be adjusted in pairs (left and right, for example) or with all four spring preload adjustments in the car to prevent crossweight changes while adjusting ride height.

TORSION BAR O.D.

Similar to an Anti-Roll Bar, a torsion bar is a spring that exerts resistive forces via applied torque generated through suspension travel. However, these torsion bars are fixed to the chassis at one end, and thus resist movement only on one wheel in the same way a coil spring resists movement and load changes. Increasing the torsion bar's diameter gives a higher spring rate, and reducing the diameter gives a lower spring rate. Stiffer springs are very helpful for smooth tracks and applications where a high level of aerodynamic attitude control is required, however stiff springs reduce mechanical grip significantly, especially over bumps. On low-grip and/or bumpy tracks, as well as lower speed tracks where aerodynamics may not be as effective, softer springs will increase mechanical grip while sacrificing aerodynamic control. Torsion Bar Diameter adjustments should be made in conjunction with ride height adjustments to prevent unwanted grounding of the chassis while on track.

LS COMP DAMPING

Low Speed Compression affects how resistant the shock is to compression (reduction in length) when the shock is moving at relatively low speeds, usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. Higher values will increase compression resistance and transfer load onto a given tire under these low-speed conditions more quickly, inducing understeer. Lower values will slow weight transfer to a tire, reducing understeer when applied to the front shocks.

HS COMP DAMPING

High Speed Compression affects the shock's behavior in high-speed travel, usually attributed to curb strikes and bumps in the track's surface. Higher compression values will cause the suspension to be stiffer in these situations, while lower values will allow the suspension to absorb these bumps better but may hurt the aerodynamic platform around the track.

HS COMP DAMPING SLOPE

The Compression Damping Slope setting controls the overall shape of the high-speed compression side of the shock. Lower slope values produce a flatter, more digressive curve while higher values result in a more linear and aggressive compression graph. The value of the slope setting is very important in controlling bump absorption at high shock velocities and controlling the aerodynamic platform. A lower slope will be helpful for rougher tracks in absorbing bumps and sharp impacts such as curbs, while a higher slope will keep the suspension more rigid. It's important to understand that these settings will affect the range the High-Speed Compression will have, with higher slope values producing a higher overall force for high-speed compression.

LS REBOUND DAMPING

Low-speed Rebound damping controls the stiffness of the shock while extending at lower speeds, typically during body movement as a result of driver inputs. Higher rebound values will resist expansion of the shock, lower values will allow the shock to extend faster. Higher rebound values can better control aerodynamic attitude but can result in the wheel being unloaded when the suspension can't expand enough to maintain proper contact with the track. When tuning for handling, higher front low-speed rebound can increase on-throttle mechanical understeer (but reduce splitter lift) while lower values will maintain front end grip longer, helping to reduce understeer, but will allow more splitter lift. Excessive front rebound can lead to unwanted oscillations due to the wheel bouncing off of the track surface instead of staying in contact.

HS REBOUND DAMPING

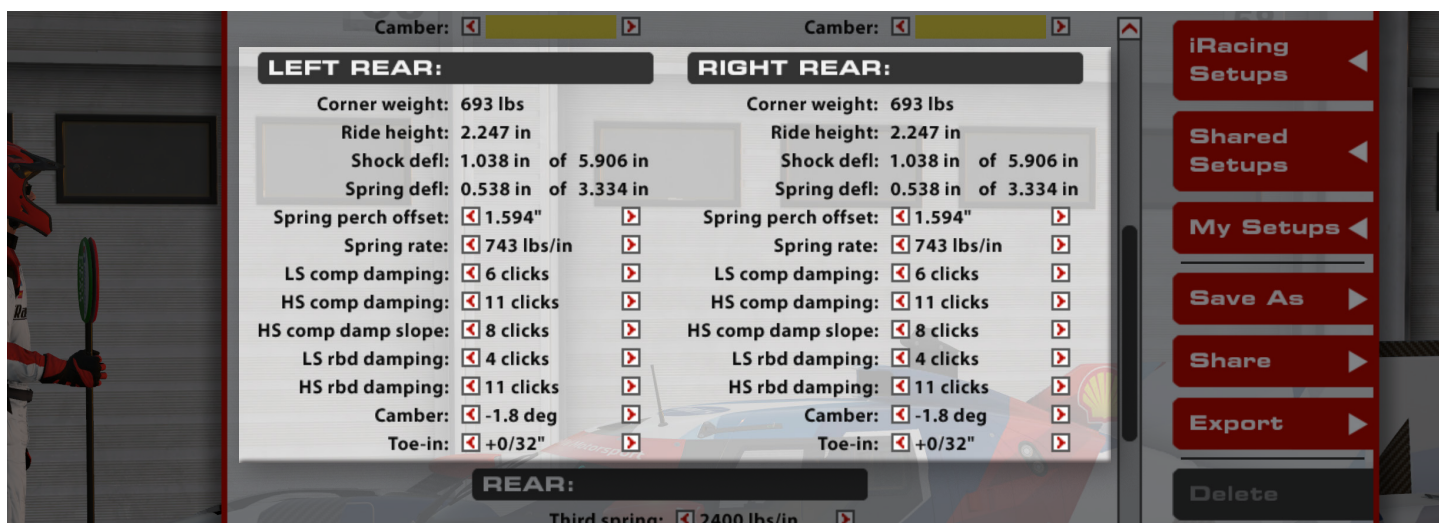
High-speed rebound adjusts the shock in extension over bumps and curb strikes. Higher values will reduce how quickly the shock will expand, while lower values will allow the shock to extend more easily. Despite not having as much of an effect on handling in result to driver inputs, High-speed rebound can produce similar results in terms of aerodynamic control and uncontrolled oscillations if set improperly.

CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Due to suspension geometry and corner loads, negative camber is desired on all four wheels. Higher negative camber values will increase the cornering force generated by the tire, but will reduce the amount of longitudinal grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance.

CHASSIS

REAR CORNERS



CORNER WEIGHT

The weight underneath each tire under static conditions in the garage. Correct weight arrangement around the car is crucial for optimizing a car for a given track and conditions. Individual wheel weight adjustments and crossweight adjustments are made via the Spring Perch Offset setting.

RIDE HEIGHT

Distance from ground to a reference point on the chassis. Since these values are measured to a specific reference point on the car, these values may not necessarily reflect the vehicle's ground clearance, but instead provide a reliable value for the height of the car off of the race track at static values. Adjusting Ride Heights is key for optimum performance, as they can directly influence the vehicle's aerodynamic performance as well as mechanical grip. Raising the rear ride height will increase overall downforce and shift aero to the front of the car but will increase drag. Decreasing rear ride height will do the opposite, with aero shifting rearward and overall downforce and drag decreasing.

SHOCK DEFLECTION

Shock Deflection is how much the shock has compressed from its fully extended length while under static conditions in the garage. This is useful for determining how much shock travel is available before a bump stop is engaged on the shock.

SPRING DEFLECTION

Spring Deflection shows how much the spring has compressed from its unloaded length. This can be used to see spring preload under static conditions and compare it against other corners of the car, with higher values representing more preload on a given spring.

SPRING PERCH OFFSET

Used to adjust ride height and corner weight, adjusting this setting applies a preload to the spring under static conditions. Decreasing the value increases preload on the spring, adding weight to its corner and increasing the ride height at that corner. Increasing the value does the opposite, reducing height and weight on a given corner. These should be adjusted in pairs (left and right, for example) or with all four spring preload adjustments in the car to prevent crossweight changes while adjusting ride height.

SPRING RATE

Spring Rate changes how stiff the spring is, represented in a force per unit of displacement. Primarily responsible for maintaining ride height and aerodynamic attitude under changing wheel loads, stiffer springs will maintain the car's aero platform better while sacrificing mechanical grip. Softer springs will deal with bumps better and increase mechanical grip, but will cause the car's aerodynamic platform to suffer. Due to homologation rules, rear spring rates must be symmetrical across the rear axle and can only be changed in pairs.

LS COMP DAMPING

Low Speed Compression affects how resistant the shock is to compression (reduction in length) when the shock is moving at relatively low speeds, usually in chassis movements as a result of driver input (steering, braking, & throttle) and cornering forces. Higher values will increase compression resistance and transfer load onto a given tire under these low-speed conditions more quickly, inducing understeer on throttle application.

HS COMP DAMPING

High Speed Compression affects the shock's behavior in high-speed travel, usually attributed to curb strikes and bumps in the track's surface. Higher compression values will cause the suspension to be stiffer in these situations, while lower values will allow the suspension to absorb these bumps better but may hurt the aerodynamic platform around the track.

HS COMP DAMPING SLOPE

The Compression Damping Slope setting controls the overall shape of the high-speed compression side of the shock. Lower slope values produce a flatter, more digressive curve while higher values result in a more linear and aggressive compression graph. The value of the slope setting is very important in controlling bump absorption at high shock velocities and controlling the aerodynamic platform. A lower slope will be helpful for rougher tracks in absorbing bumps and sharp impacts such as curbs, while a higher slope will keep the suspension more rigid. It's important to understand that these settings will affect the range the High-Speed Compression will have, with higher slope values producing a higher overall force for high-speed compression.

LS REBOUND DAMPING

Low-speed Rebound damping controls the stiffness of the shock while extending at lower speeds, typically during body movement as a result of driver inputs. Higher rebound values will resist expansion of the shock, lower values will allow the shock to extend faster. Higher rebound values can better control aerodynamic attitude but can result in the wheel being unloaded when the suspension can't expand enough to maintain proper contact with the track. When tuning for handling, higher rear low-speed rebound can increase off-throttle mechanical understeer (but reduce rear-end lift) while lower values will maintain rear end grip longer, helping to reduce oversteer, but will allow more rear end lift under deceleration. Excessive rear rebound can lead to unwanted oscillations due to the wheel bouncing off of the track surface instead of staying in contact.

HS REBOUND DAMPING

High-speed rebound adjusts the shock in extension over bumps and curb strikes. Higher values will reduce how quickly the shock will expand, while lower values will allow the shock to extend more easily. Despite not having as much of an effect on handling in result to driver inputs, High-speed rebound can produce similar results in terms of aerodynamic control and uncontrolled oscillations if set improperly.

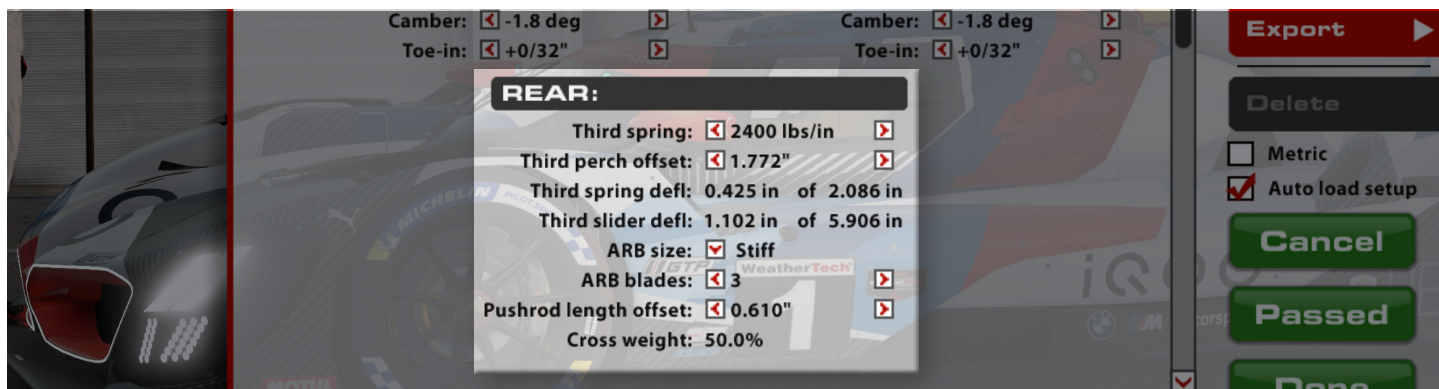
CAMBER

Camber is the vertical angle of the wheel relative to the center of the chassis. Negative camber is when the top of the wheel is closer to the chassis centerline than the bottom of the wheel, positive camber is when the top of the tire is farther out than the bottom. Due to suspension geometry and corner loads, negative camber is desired on all four wheels. Higher negative camber values will increase the cornering force generated by the tire, but will reduce the amount of grip the tire will have under braking. Excessive camber values can produce very high cornering forces but will also significantly reduce tire life, so it is important to find a balance between life and performance. Higher rear camber values can increase cornering stability but reduce stability under braking.

TOE-IN

Toe is the angle of the wheel, when viewed from above, relative to the centerline of the chassis. Toe-in is when the front of the wheel is closer to the centerline than the rear of the wheel, and Toe-out is the opposite. On the rear end, adding toe-in will increase straight-line stability but may hurt how well the car changes direction.

REAR



THIRD SPRING

The Third Spring, similar to the front Heave Spring, is a spring element configured to provide resistance only in vertical suspension movement without affecting roll stiffness. This spring element is helpful with controlling increasing aerodynamic loads and maintaining the proper aerodynamic attitude around a circuit.

The rear end's third spring is crucial in maintaining and controlling the rear ride height around a circuit to maximize the downforce produced by the rear bodywork.

THIRD PERCH OFFSET

The Third Perch Offset is used to adjust preload on the rear Third Spring. This is one of two methods to adjust ride height through the rear Third Spring element, with lower values preloading the spring more and raising front ride heights. Conversely, higher values will unload the spring and lower front ride heights.

THIRD SPRING DEFLECTION

Third Spring Deflection represents the amount the rear Third Spring is compressed under static conditions. This is not directly adjustable but will change with adjustments to the Third Perch Offset and rear Spring settings.

THIRD SLIDER DEFLECTION

The Slider Deflection is how far the slider mechanism the Third Spring is mounted on has compressed from fully extended. Similar to a shock but without any damping forces produced, this doesn't influence the suspension's behavior.

ARB SIZE

The ARB (Anti-Roll Bar) size alters the stiffness of the rear suspension in roll. Increasing the ARB size will increase the roll stiffness of the rear suspension, resulting in less body roll but increasing mechanical oversteer. Conversely, reducing the ARB size will soften the suspension in roll, increasing body roll but decreasing mechanical oversteer. This can result in a less-responsive feel from the steering, but grip across the rear axle will increase.

ARB BLADES

The configuration of the Anti-Roll Bar arms, or "blades", can be changed to alter the overall stiffness of the ARB assembly. Higher values transfer more force through the arms to the ARB itself, increasing roll stiffness in the rear suspension and producing the same effects, albeit on a smaller scale, as increasing the diameter of the Anti-roll bar. Conversely, lower values reduce the roll stiffness of the rear suspension and produce the same effects as decreasing the diameter of the Anti-roll bar. These blade adjustments can be thought of as fine-tuning adjustments between Anti-roll bar diameter settings.

BRAKES/DRIVE UNIT

LIGHTING & BRAKE SPEC



ROOF ID LIGHT COLOR

The color of the identifier lights on the roof of the car can be changed to better identify similarly-painted cars in nighttime conditions. This adjustment has no effect on vehicle performance.

PAD COMPOUND

The vehicle's braking performance can be altered via the Brake Pad Compound. The "Low" setting provides the least friction, reducing the effectiveness of the brakes but allowing for better brake pressure modulation, while "Medium" and "High" provide more friction and increase the effectiveness of the brakes while increasing the risk of a brake lockup.

FRONT MASTER CYLINDER

The Front Brake Master Cylinder size can be changed to alter the line pressure to the front brake calipers. A larger master cylinder will reduce the line pressure to the front brakes, which will shift the brake bias rearwards and increase the pedal effort required to lock the front wheels. A smaller master cylinder will increase brake line pressure to the front brakes, shifting brake bias forward and reducing required pedal effort to lock the front wheels.

REAR MASTER CYLINDER

The Rear Brake Master Cylinder size can be changed to alter the line pressure to the rear brake calipers. A larger master cylinder will reduce the line pressure to the rear brakes, which will shift the brake bias forwards and increase the pedal effort required to lock the rear wheels. A smaller master cylinder will increase brake line pressure to the rear brakes, shifting brake bias rearward and reducing required pedal effort to lock the rear wheels.

BRAKE PRESSURE BIAS

Brake Bias is the percentage of braking force that is being sent to the front brakes. Values above 50% result in more pressure being sent to the front, while values less than 50% send more force to the rear. This should be tuned for both driver preference and track conditions to get the optimum braking performance for a given situation.

HYBRID CONFIG & FUEL



The Hybrid power system on the BMW LMDh can be set to one of five modes to change the target battery State of Charge (SoC) after each lap. Each of these modes will use varying levels of energy throughout a lap to reach a target, and thus some will produce more power over the course of a lap and faster lap times at the cost of discharging the battery.

NO DEPLOY

In the "No Deploy" mode, the Hybrid system will not use any energy stored in the battery. This essentially disables the Hybrid drive system and will only charge the battery throughout a lap. This is only available in Qualifying and Test sessions and is used to fully charge the battery before switching to Qual mode.

QUAL

This mode is intended to be used on flying laps during qualifying sessions and will attempt to use all of the battery charge during a lap. This is only available during Qualifying and Test sessions and should be preceded by the No Deploy setting on outlaps and warmup laps to ensure the battery is fully charged before switching to the Qual mode.

ATTACK

Attack mode reduces the target State of Charge to use more power during race sessions to help with overtaking. Generally the laptime gain from this mode is not enough to offset the loss in pace from having to recharge and recover from using Attack mode, so it should be used only when it is absolutely necessary to complete an overtake. This mode can also be used on the final lap for a burst of speed since the battery is no longer needed. This mode is only available for Practice, Race, and Test sessions.

BALANCED

The Balanced mode is the primary Race mode for the Hybrid system. This mode will attempt to deploy electrical charge to reduce lap times as much as possible while still maintaining a reasonable State of Charge over the duration of a lap. At the start of a session, it will take a few flying laps for the Hybrid system to learn the track and optimize deployment for the best lap times, and this mode is only available in Practice, Race, and Test sessions.

BUILD

The Build mode will attempt to build battery charge as quickly as possible in the event of a low battery charge or if it is needed prior to switching to Attack mode. Note that this will compromise lap times significantly compared to Balanced, and it's important to switch back to Balanced mode once the battery has charged to avoid losing harvested energy and to prevent unnecessary loss in pace. This mode is only available in Practice, Race, and Test sessions.

FUEL LEVEL

Fuel level is the amount of fuel in the fuel tank when the car leaves the garage.

TRACTION CONTROL, GEAR RATIOS, & REAR DIFF SPEC



TRACTION CONTROL GAIN

Gain is the amount of intervention the Traction Control will exert when wheel spin is detected. Higher values result in a more aggressive throttle cut to control wheelspin.

TRACTION CONTROL SLIP

Slip is how sensitive the Traction Control system will be to wheelspin. Higher values will activate the Traction Control system with smaller amounts of wheelspin, while lower values will allow slightly more wheelspin prior to activating the system.

GEAR STACK

Gear Stack changes the gear ratios in the transmission. Two choices are available: Short and Long. The Short setting will choose a more acceleration-focused gear set for tracks with shorter straights or slower corners, while the Long option will choose gears more suited to high-speed tracks with long straights.

GEAR SPEEDS

Each of the transmission's seven forward gears will show the approximate ground speed at which the engine will reach maximum RPM. These values will change based on which Gear Stack is selected, but the true maximum speed may differ slightly due to on-track conditions.

COAST/DRIVE RAMP ANGLES

Coast and Drive Ramp Angles affect the force exerted by the differential to keep both driven tires locked together under acceleration. Lower values produce more locking force, and more locking force increases understeer during braking and acceleration phases. Higher values will produce less locking force and induce oversteer in these situations.

CLUTCH FRICTION PLATES

The number of clutch faces affect how much overall force is applied to keep the differential locked. Treated as a multiplier, adding more faces produces increasingly more locking force.

PRELOAD

The differential can be set with a static load applied. Higher values produce more locking force in the differential in all conditions, producing more understeer under acceleration and deceleration. This value will also affect mid-corner performance, with higher values not allowing the differential to unlock as much, increasing mid-corner understeer.

SETUP TIPS

This section is aimed toward helping users who want to dive deeper into the different aspects of the vehicle's setup.

SETUP TIPS

PROVIDED SETUPS

There are 12 iRacing setups provided for the BMW M Hybrid V8 LMDh:

BASELINE

A slightly more stable version of the high downforce setup for a driver's first time in the car or for those wanting more stability than the high downforce setup provides.

HIGH DOWNFORCE

For use at most tracks. Although quite efficient and creating the most downforce, this setup also makes the most drag.

MEDIUM DOWNFORCE

For use at track with long straights like Spa. Sacrificing more downforce and efficiency for more straight line speed.

LOW DOWNFORCE

For use at Le Mans. Lowest drag and downforce trim.

DAYTONA

A variation of the Low Downforce setup for use at Daytona with changes to avoid bottoming on the oval banking.

DAYTONA 24H

A variation on the Daytona setup made specifically with time of year, expected weather, and the demands of a 24 hour race setup in mind.

FIXED

The high downforce setup with limited fuel for the official IMSA fixed series.

FIXED QUAL

The fixed series setup with qualifying fuel and hybrid deployment.

FIXED LEMANS

The low downforce setup with limited fuel for the official IMSA fixed series.

FIXED LEMANS QUAL

The fixed LDF series setup with qualifying fuel and hybrid deployment.

FIXED DAYTONA

The Daytona setup with limited fuel for the official IMSA fixed series.

FIXED DAYTONA QUAL

The Daytona series setup with qualifying fuel and hybrid deployment.

The fuel level in all setups (except the fixed IMSA setups) are set to 89 L, which is the maximum tank capacity for this car.

CHASSIS HEIGHTS

The BMW M Hybrid V8 LMDh generates the most downforce (and is generally the most efficient) with an average rear ride height of 50mm and an average front ride height as low as possible. It's important to note that the least amount of drag is generated when both ends of the car are as low as possible (without bottoming).

More rake (high rear ride heights compared to front ride heights) will move aero balance forward (oversteer) and less rake will move it rearwards (understeer).

You can affect the dynamic ride heights (without changing the roll stiffness) on track by adjusting the front heave spring and the rear third spring stiffnesses.

In general the front heave spring should be as soft as possible to allow the nose of the car to be as low as possible without excessive bottoming or an undesirable aero balance shift under braking/turn in.

A soft rear third spring will allow the rear of the car to drop under aero load reducing drag but you will lose some amount of downforce and efficiency mid corner as the rear ride heights will be well under the target rear heights for maximum efficiency (50mm). Some compromises will need to be made regarding drag/downforce and dynamic aero balance.

AERODYNAMIC ADJUSTMENTS

Should you choose to make adjustments to the setup, the easiest way to change the aerodynamic balance of the setup is through adjustment of the rear wing angle. Generally speaking, if you find you want to adjust wing position in more than a click in either direction it is recommended you start from one of the other downforce trim setups (high, medium, or low).

Lower wing angle = More oversteer, less downforce, and higher straight line speed.

Higher wing angle = More understeer, more downforce, and lower straight line speed.

If you would like to change the aero balance without changing the downforce/drag trim adjust the rear ride height by increasing or decreasing the rear pushrod length offset. This will affect balance in all situations due to the high amount of downforce but this will be particularly noticeable in mid and high speed corners.

Lower pushrod length delta = Lower rear ride height, less rake, and a more rearward aero balance (understeer).

Higher pushrod length delta = Higher rear ride height, more rake, and a more forward aero balance (oversteer).

MECHANICAL ADJUSTMENTS

If you would like to change the mechanical balance without changing the aero balance the front and rear ARBs and ARB blades may be adjusted. This can be particularly useful if you are happy with the general balance of the car but want either more or less rotation in slow speed corners.

Bigger front ARB/higher blade value = More understeer

Smaller front ARB/lower blade values = Less understeer

Bigger rear ARB/higher blade values = More oversteer

Smaller rear ARB/lower blade values = Less oversteer

DIFFERENTIAL ADJUSTMENTS

The rear differential is a powerful tool in the Acura, and the simplest change one can make is increasing or decreasing the preload. While adjusting the preload will affect the on throttle behavior you will probably notice more of a change on entry and turn in.

More preload = Less rotation off throttle, more stability on entry.

Less preload = More rotation off throttle, less stability on entry.